

# Land Allocation Modeling with New Theory Principles

## Summary Sheet (MMC051)

July 19, 2026

### Problem Statement and Background Information

The "New Theory" is a concept introduced by King Rama IX of Thailand emphasizing self-efficiency for farmers. The prominent recommendation for dividing standard 1-rai ( $1600\text{ m}^2$ ) land is a 30 : 30 : 30 : 10 ratio for water resources, rice farmings, mixed crops, and residentials & utilities. The land is assumed to be uniformly flat, square, and receives equal sun in all directions, however; real-world land plots are frequently imperfect and irregular, with different elevations and water source locations both contribute to different optimal land division.

In the real-world context, this is a major problem for farmers, as the instructed ratios may not be suitable for their land shape, nor it is guaranteed to be effective for farmers. Farmers are told to use it as a guiding principle, but farmers may not be able to adapt effectively without prior information in other fields and conditions. Therefore, our task is then to develop and suggest new composition(s) for an arbitrary quadrilateral-shaped land, and propose a flexible model that reflects the sustainability and suitability in accordance to the "New Theory", and it should be easily adapted by the farmers.

### Key Assumptions

The following assumptions are to be made, as per the problem statement:

- The area of the land is 1-rai or  $1,600\text{ m}^2$
- The shape of the land is an asymmetrical quadrilateral
- The elevation of the land slopes downward, from westernmost highland to lowland in the east.
- The pond should have sufficient water for year-around agricultural use.
- The residential & utility zone must be located on the highland and connected to the roadway.

Moreover, in these model(s), we add these additional assumptions:

- Assume that the incline of elevation is constant
- Assume that a single roadway is located adjacent to one side of the land in parallel with its border.
- Assume that the quadrilateral is convex.
- Assume that the rice is cultivated once a year from July to October, when the rainy season lies.

### Model Overview

We let the land be divided into 4 zones: water resources (zone 1), rice farming (zone 2), mixed crops cultivation (zone 3), and residential & utility (zone 4).  $A_i$  is the area ( $\text{m}^2$ ) of the  $i^{\text{th}}$  zone.  $W_i$  is the weighting coefficient, indicating the importance of the  $i^{\text{th}}$  zone. We consider the conditions that benefit the zone, such as the elevation, closeness to water source, or sufficient sunlight, as efficiency, denoted as  $E_i$ . The overview model is shown below.

$$\text{Total Efficiency} = A_1W_1E_1 + A_2W_2E_2 + A_3W_3E_3 + A_4W_4E_4$$

The model first fix the ratio  $A_1 : A_2 : A_3 : A_4$ . Then it begins to formulate the land allocation which outputs the highest total efficiency. Subsequently, the program iteratively change the initial ratios and compute

again. The computation is done with a generic square grid ( $40\text{ m} \times 40\text{ m}$ ), and an example asymmetrical quadrilateral for comparison.

## Core Results

For both a generic square grid ( $40\text{ m} \times 40\text{ m}$ ), and an asymmetrical quadrilateral, with the inclination of  $10^\circ$ , the optimal ratio between water source : rice farm : mixed crop : resident =  $50 : 20 : 20 : 10$ . Then the efficiency drops greatly and ties between ratios  $40 : 30 : 20 : 10$  and  $40 : 20 : 20 : 20$ . Therefore, we propose the  $50 : 20 : 20 : 10$  ratio to be a flexible ratio in accordance to the core philosophy of the "New Theory", prioritizing self-sustenance and suitability, based on the example results.

## Model Assessment

For suitability, the model considered the shape of the water resource to be a frustum which reflect a commonly used ratio for pond shaping. It also allocates the rice field and mixed crops to be close to the water for easy access and lowest cost in pipeline system implementation, ensuring all plants are hydrated properly. Subsequently, the rice field is placed lower than the mixed crops, effectively reducing frequent water replenishments. Lastly, the residential area will always have direct access to the road, and is at a high elevation as protection from floods.

For sustainability, water sustainability is at the center of our attention. Farmers should have sufficient water that lasts a certain period in dry seasons or climate changes. Nonetheless, the rice field must also cover daily food consumptions of the farmers. We constrained the rice yield to exceed the average amount of rice consumed yearly by Thais, as reflected in water usage constants. And despite the sloped terrain, we have also modeled a simple method in determining sunlight blockage from the sun declination and the position of the residential area (supposed highest elevation) to ensure enough sunlight throughout.

## Recommendations and Limitations

As our model has several assumptions regarding the water usage, basic needs, and farming duration, the outcome only suggests the optimal ratio of land for those assumptions derived from researched statistics. As a result, the model must be initialized with those parameters according to the user's criterias first.

According to the New Theory, our model prioritizes only the essentials and water resources in case of emergency, not aiming to profit from any excess rice or crops. Thereby, if farmers aim to do so, which is also practical, they should increase the weight of rice and crop yields according to their needs.

## Utilization of the Results

Our results compose of both the area ratios of the zones, and a preliminary diagram of the land allocations. The farmers that use our model are able to make adjustments according to their own preferences, this may include,

- Constructing canals around the land for efficient water translocation around the land.
- Change the shape of each zone to match their usages, while splitting zones based on the resulting center of masses, the "center" point.
- Connect pathways or other residential areas across other zones.
- Connect disconnected or adjust narrow pathways for convenience.

Hence, our results are convenience for the farmers, easy to understand, and can be implemented across various plots of land easily and fast.